1 What the strategies do

Two builds of the same NQ day-hold entry. The BASE build is the entry and the timed exit alone. The FILTERED build is the same entry and exit with two conditions on the day that must hold before the signal is allowed to fire. Neither build has a stop, a target or a trail.

1.1 The signal

The reference bar is the first five-minute bar of the New York session, the one that closes at 08:35 Chicago time. At that bar's close the signal compares the close to a 12-bar EMA of five-minute closes. Close above the EMA buys, close below sells short, at that bar's close. One signal per calendar day.

The trade is a bet that the direction of the opening five minutes persists into the close. It is held for the whole session whatever happens in between.

1.2 The exit

The position is closed at the close of the first bar at or after 15:00 Chicago time, the 16:00 New York cash close. `SetExitOnClose` is set as a backstop, so on a half day where the 15:00 bar never prints the session end closes the trade instead.

1.3 The filters (FILTERED build only)

Both conditions are measured in units of the average day-session range, the mean high-low of the last `ADRLength` day sessions from 08:30 to 15:00. A range unit is used instead of points so that a setting means the same at NQ 7,000 as at NQ 25,000.

- **Reference bar strength.** The first five-minute bar's own range must be at least `RefBarMinRange` of the average day range. The filter asks for a real opening move before the direction of that move is trusted.
- **Volatility regime.** The 20-session average day range must be at least `VolRegimeRatio` times the 100-session average. The entry only trades when intraday ranges are expanding against their own recent history.

A day that fails either test takes no trade, and there is no second attempt that day.

> **The regime filter needs history.** No trade is taken until `VolRegimeLongLen` sessions (100 by default, about five months) have been stored. Backtests of the FILTERED build start later than the BASE build on the same data; compare the two from the same date.

2 Setting it up

Both signals run on one data series and need nothing beyond the inputs. The chart they expect:

Group	NQ opening-direction day hold
Instrument	NQ continuous contract (@NQ)
Data 1	5 Min, 24-hour Globex session
Data 2	none
Time zone	CT. Defaults are set for an Exchange (Chicago) chart.

Step 1. Build the chart. NQ, five-minute bars, full Globex session, chart time zone Exchange (Chicago). The signal reads clock times, so a chart in another time zone shifts the reference bar and the exit without any error.

Step 2. Bars back. Auto-detect is enough; nothing looks back further than the EMA.

Step 3. Load the signal. Compile `NQ_First5_EMA_DayHold_BASE` or `NQ_First5_EMA_DayHold_FILTERED`, add one of them to the chart, and check that the input list matches section 3.

Step 4. Costs and fills. Set commission and slippage in the strategy properties. Bar Magnifier is not needed, since every order fills at a bar close.

> **Changing the bar size changes the reference bar.** The signal is defined on the first five-minute bar. On one-minute bars `SignalTime` 0835 is the fifth bar of the session and the EMA is a different average; on fifteen-minute bars the reference bar does not exist. Keep the chart at five minutes.

3 The inputs

These are the defaults in both builds. The BASE build has the first six; the FILTERED build adds the rest.

3.1 Entry and exit (both builds)

INPUT	DEFAULT	WHAT IT DOES
`EMALength`	`12`	Length of the EMA the reference bar close is compared to.
`SignalTime`	`0835`	Close time of the reference bar, Chicago time.
`ExitTime`	`1500`	The first bar closing at or after this time exits at its close. 1500 CT is the 16:00 ET cash close.
`AllowLongs`	`true`	false disables long entries.
`AllowShorts`	`true`	false disables short entries.
`NumContracts`	`1`	Contracts per entry.

3.2 Filters (FILTERED build)

INPUT	DEFAULT	WHAT IT DOES
`SessionOpen`	`0830`	Day-session open, Chicago time. Bars after this and up to ExitTime build the day range.
`RefBarMinRange`	`0.15`	Minimum range of the reference bar as a fraction of the average day range. 0 switches the filter off.
`VolRegimeRatio`	`1.10`	The 20-session average day range must be at least this multiple of the 100-session average.
`ADRLength`	`20`	Sessions in the short average day range.
`VolRegimeLongLen`	`100`	Sessions in the long average day range. Also the warm-up length.

`VolRegimeRatio` is the input that decides how many days trade: raising it admits fewer days, all of them in higher-volatility stretches. `RefBarMinRange` removes the days where the opening bar itself was too small to carry a direction.